

Simplifying $\gcd(\text{lcm}(a,b), \text{lcm}(b,c)) = \gcd(a,c)$

Step 1: Prime Exponent Representation

For an arbitrary prime p , let:

$$v_p(a)=x, \quad v_p(b)=y, \quad v_p(c)=z$$

Equivalently, with respect to the prime p :

a contains p^x , b contains p^y , c contains p^z .

LCM

$$\text{lcm}(a,b) = p^{\max(x,y)}$$

$$\text{lcm}(b,c) = p^{\max(y,z)}$$

GCD of the LCMs

$$\begin{aligned} &\gcd(\text{lcm}(a,b), \text{lcm}(b,c)) \\ &= p^{\min(\max(x,y), \max(y,z))} \end{aligned}$$

Right-hand Side

$$\begin{aligned} &\gcd(a,c) \\ &= p^{\min(x,z)} \end{aligned}$$

Therefore the exponent equation becomes:

$$\min(\max(x,y), \max(y,z)) = \min(x,z)$$

Step 2: Apply the Identity

Use the identity:

$$\begin{aligned} \min(\max(x,y), \max(y,z)) \\ = \\ \max(y, \min(x,z)) \end{aligned}$$

Hence,

$$\begin{aligned} \max(y, \min(x,z)) \\ = \\ \min(x,z) \end{aligned}$$

Let

$$t = \min(x,z)$$

Then

$$\max(y,t)=t$$

This is true iff

$$y \leq t$$

or

$$y \leq \min(x, z)$$

Final Condition

For every prime p ,

$$v_p(b) \leq v_p(a)$$

and

$$v_p(b) \leq v_p(c)$$

This is exactly equivalent to:

b divides a and b divides c .

Final Simplification

$$\gcd(\text{lcm}(a, b), \text{lcm}(b, c)) = \gcd(a, c)$$

holds if and only if

$b \mid a$ and $b \mid c$.